Tutorial 4 – Key Issues in Software Design

1. Describe what concurrency in software development is.
2. How do we characterize concurrency?
3. What are the challenges in dealing with concurrency in software engineering?
4. Controlling and handling the events is one of the important issues when designing software. Distinguish between *implicit invocation* and *callback* mechanisms that are used in the controlling and handling of the events process.

5. Discuss what is meant by middleware.

6. Discuss different types of middleware in details.

7. Using an example of a remote procedure call (RPC), explain how middleware coordinates the interaction of computers in a distributed system.

8. Compare exception handling with fault tolerance.

9. Security is one of the key issues that have to be dealt when designing software. Suggest **THREE (3)** types of security that are required to prevent unauthorised access to information and other resources.

10. Why is assuring system security a more challenging problem than assuring system safety?

11. **Question 2 (past mid term)**

A traffic system is proposed for a warehouse that will use robots to carry items from one location to another location. There are 4 main tracks (A, B, C and D), each supporting one robot moving back and forth. The tracks are built as shown in Figure 1. Each track allows robots to move to the designated location. The control of the traffic movement is crucial to avoid collisions between the robots at the intersection. For example, if a robot is allowed to move on track A, robot on parallel track B can also move at the same time. Robots on the intersecting tracks C and D must be stopped for 1 minute. Turnings to another track are not allowed (i.e. robot on Track A cannot turn onto Track C/D). Before a track is opened for movement, sensors will send signals to traffic light to issue stop signals to robots. The warehouse owners are willing to allocate extensive resources for the development of the system to achieve their goals.

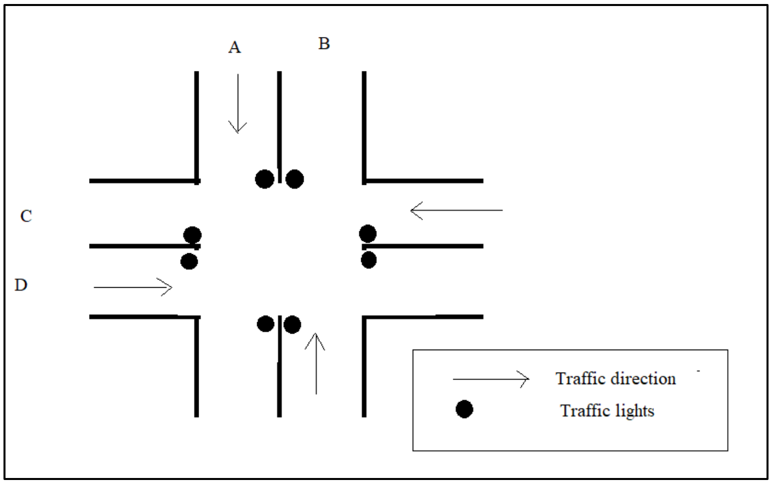

Figure 1: Warehouse traffic system for robots

Use your own word to resolve the following questions:

a) Identify and explain the **THREE (3)** key issues in this traffic system. (6 marks)

b) Examine **THREE (3)** detailed design qualities that are crucial for the system. (6 marks)

c) In designing the system, examine how middleware will help to reduce the consumption of system development resources. Provide **THREE (3)** distinct answers. (9 marks)

d) Explain **TWO (2)** concerns of the system by the owners. (4 marks)

[Total: 25 marks]

Additional questions

a) Differentiate *Fault Tolerance* with *Exception Handling*.

b) Analyse **FOUR (4)** security approaches that you need to implement into an online ticketing system when designing the system.

c) Explain the term Middleware.

d) A system is proposed to monitor the water level of Africa's rivers for the possibility of floods that may happen. For each river, the system detects the water level and its flow rate through motion sensors that are attached to a small processing unit located along the river bank. The distance between sensors is around 800 meters.

(i) Examine **THREE (3)** safety or security issues that the designer needs to considered.

ii) Analyse. the importance of fault tolerance on the system.

Activity Based Question

1. Perform a finding on con-currency in software design. Explain the effects of concurrency to the coding of software with examples.

2. Access to the url: https://www.codeproject.com/Articles/16349/Managing-a-Concurrent-System-Design. Convey your understanding through a presentation to the class.